Paleomagnetic study of Siluro-Devonian volcanic rocks from the central Lachlan Orogen: Implications for the apparent pole wander path of Gondwana

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[1] The apparent pole wander (APW) path for Gondwana is still not clearly established, in particular, for Silurian-Devonian times. A controversial debate places authors who argue for an "X path," running directly through Africa on a reconstruction of Gondwana against those who advocate a large loop passing by southern South America, the "Y path." Most of the paleomagnetic data used to draw this loop come from the Lachlan Orogen (Australia). A paleomagnetic study was carried out in the well-dated Ambone and Ural volcanics in the central subprovince of Lachlan Orogen, New South Wales. Anisotropy of magnetic susceptibility measurements confirms detailed mapping of the region and shows that these massive dacitic sills and/or lava flows are flat lying. Among the different localities studied, only one yields interpretable paleomagnetic results. Two components of magnetization can be identified: a midtemperature direction yielding a corresponding pole in Australian coordinates $\lambda = 67.9°S/\phi = 084.4°E$ (B = 5; n = 21; dp = 17.5°/dm = 23.1°) and a high-temperature direction with a corresponding VGP $\lambda = 24.4°S/\phi = 060.6°E$ (B = 5; n = 25; dp = 1.4°/dm = 2.5°). The first is interpreted as corresponding to an Early Carboniferous pole position and can be regarded as an overprint probably related to the Early Carboniferous Kanimblan orogenic event. The second does not correspond to any expected Silurian-Devonian or younger pole position. This magnetization is thought to be primary in origin; however, secular variation has apparently not been averaged out in the single lava flow sampled. Therefore the earliest Devonian paleopole position probably lies in a 30° cone around the obtained VGP, and this position can only match the X-type APW path for Gondwana. It is in particular very different from coeval poles obtained in the eastern subprovince of the Lachlan Orogen, and it is mostly used as key poles supporting the Silurian-Devonian loop for the APW path of Gondwana. Therefore some poles from the Lachlan Orogen must be affected by rotation to explain these results. The Lachlan Orogen was hence not stable up to the mid-Paleozoic, and data from this region should not be used as representative for Gondwana.

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1. Introduction

[2] Although many apparent polar wander (APW) paths for Gondwana have been published in the last 30 years, reconstructing the paleogeographic movements, over time, of this huge landmass that comprises about the two thirds of the present-day continent, still remains a problem. The paleoposition and configuration of Gondwana has broad repercussions for tectonics, paleoclimates and paleobiogeography.

[3] There is consensus in the literature (e.g., see following references) that the Cambro-Ordovician paleopole for Gondwana lies in the vicinity of Morocco (on a reconstruction of Gondwana in African coordinates) and that the Late Carboniferous to Permian path passes through Antarctica (Figure 1). The paleopole positions for the mid-Paleozoic times, are, however, more problematical. There are basically two schools of thought: authors who draw the APW path more or less directly through Africa (e.g., X path of *Morel and Irving* [1978], *Bachtadse and Briden* [1991], *Grunow* [1999], *Scotese et al.* [1999], and *Smith* [1999]), and those who involve a Silurian-Devonian loop passing by southern South America (e.g., Y path of *Morel and Irving* [1978], *Bachtadse and Briden* [1990], *Kent and Van der Voo* [1990], *Schmidt et al.* [1990], *Li and Powell* [2001], *Torsvik and Van der Voo* [2002], and *McElhinny et al.* [2003]). The

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Y type of APW path implies very high drift rate (>30 cm/yr), in particular for a landmass as huge as Gondwana [see, e.g., *Bachtadse and Briden*, 1990].

[4] Most of the paleomagnetic data used to draw this loop come from eastern Australia, and in particular, from the Lachlan Orogen. However, with increasing recognition that faults in the Lachlan Orogen have thrust and/or strike-slip displacement, often of unknown magnitudes, there is a realization that different blocks might have undergone rotation and translation between the Late Ordovician and the Early Carboniferous relative to the craton. In particular, blocks situated in the eastern subprovince of the Lachlan Orogen might have been amalgamated to the protomargin of Gondwana, consecutive to the period of subduction, which occurred in the Early to Mid-Paleozoic and is now marked by the Macquarie Volcanic Arc (Figure 2) (see, e.g., *Glen* [1992] and *Gray et al.* [1997] and also *Schmidt et al.* [1986, 1987] and *Morel and Irving* [1978] for an alternative model). In such a situation, the use of paleomagnetic data from Lachlan Orogen should not be used to define the APW path of Gondwana. As part of a project on the Gondwana APW path, this paper reports findings of a paleomagnetic study of rift basins in the central subprovince of the Lachlan Orogen (New South Wales) that has implications for both global paleogeography and the tectonic evolution of the orogen (Figure 2).

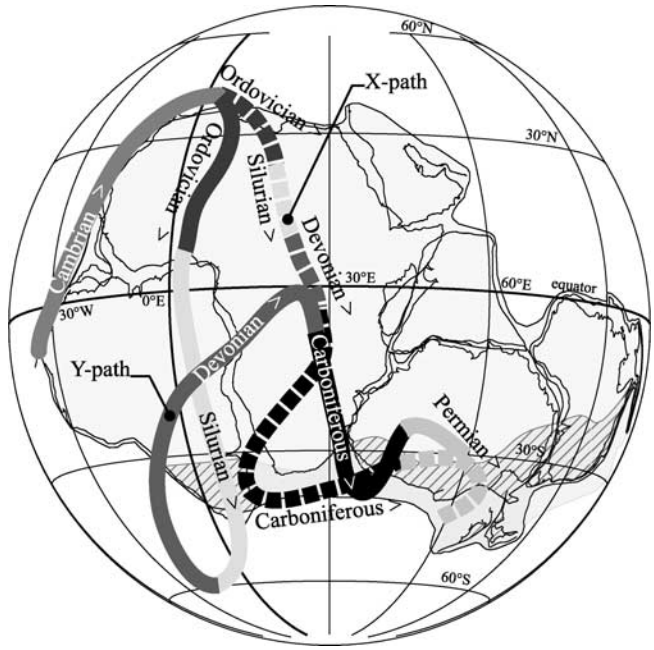


Figure 1. Two schools of thought about the shape of the APW path for Gondwana. The X path (dashed line), here after *Bachtadse and Briden* [1991], runs more or less directly through Africa during the mid-Paleozoic. The Y path (solid line), after *Schmidt et al.* [1990], involves a Silurian-Devonian loop mainly based on paleomagnetic data from the Lachlan Orogen. This orogen was referred to as part of a "suspect region" (hatching zone) according to *Van der Voo* [1993]. Reconstruction of Gondwana is in African coordinates, orthogonal projection.

2. Geological Setting

[5] The Lachlan Orogen is divided into several structural subprovinces, three in the models of *Gray et al.* [1997] and four in the models of *Glen* [1992, 2005]. Samples described in this paper are located in the central subprovince of the Lachlan Orogen, lying inboard (west) of relics of former intraoceanic island arc, the Macquarie Volcanic Arc (Figure 2) that formed from west dipping subduction, throughout the Ordovician [*Glen et al.*, 1998, 2003; *Glen*, 2004; *Glen et al.*, 2004].

[6] After the Late Ordovician–Early Silurian Benambran Orogeny in which Ordovician craton-derived turbidite terranes were assembled adjacent to the arc [*Glen*, 2005], the plate boundary between the proto-Pacific plate and the Gondwana margin underwent major rollback back to the east. As a result, the Silurian–Middle Devonian history of the Lachlan Orogen is marked by extension in a back arc position. This Tabberabberan cycle was marked by formation of rift basins and emplacement of granitic bodies [*Glen*, 2004, 2005]. Major Late Silurian–Early Devonian rift basins in the central subprovince of the Lachlan Orogen include the Cobar Basin, the Mount Hope Trough and the Rast Trough [*Glen et al.*, 1996], which were separated or flanked by shallow water shelves: the Winduck, Walters Range and Kopyje shelves (Figure 2). Overlying these were late Early to Late Devonian fluviatile strata of the Mulga Downs and Cocoparra groups [*Colquhoun et al.*, 2000]. The Mount Hope Trough [*Scheibner*, 1987] is filled by volcanics and volcaniclastic rocks of the poorly dated Late Silurian to Early Devonian Mount Hope Group that interfingers at the top with siliciclastic turbidites of the Broken Range Group [*Scheibner*, 1987]. One major unit, the Ambone Volcanics, consists of two massive lava flows separated by partly turbiditic sediments. Lava units show well-developed columnar jointing. These rocks are rhyodacite porphyries, and are thought to have been extruded as extensive submarine lava flows in the latest Pragian [*Scheibner*, 1987]. They were deformed in the late Early–Middle Devonian with formation of north-south folds with wavelengths of ~1 km. Late Devonian strata to the west and south of the region were folded into open broad folds in the Carboniferous [*Scheibner*, 1987; *Glen et al.*, 1996]. These lava flows were sampled over several hundreds of meters of section close to the Mount Hope township (termed locality "HOP"). The sampled locality was on the eastern limb of a 1-km-wide syncline occupied by Ambone Volcanics. The western limb of the structure is cleaved and sheared and was not sampled. Bedding measurements in bedded volcaniclastics of the Mount Halfway Volcanics below the Ambone Volcanics show that the columns in the volcanics are precisely perpendicular to the paleohorizontal and the northern part of the fold can be regarded as cylindrical with a horizontal axis.

[7] The Rast Trough is approximately coeval with the Mount Hope Trough, separated from it by the shallow water Walters Range shelf (Figure 2), which is inferred to have developed as a sag phase feature above basement rocks [*Glen et al.*, 1996]. Recent mapping shows that the southern part of the trough is filled mainly by the Siluro-Devonian Ural Volcanics interpreted to be rift stage fill [*Colquhoun et al.*, 2000]. Geochemical data show the volcanics are dominantly rhyolitic to dacitic in composition but range to

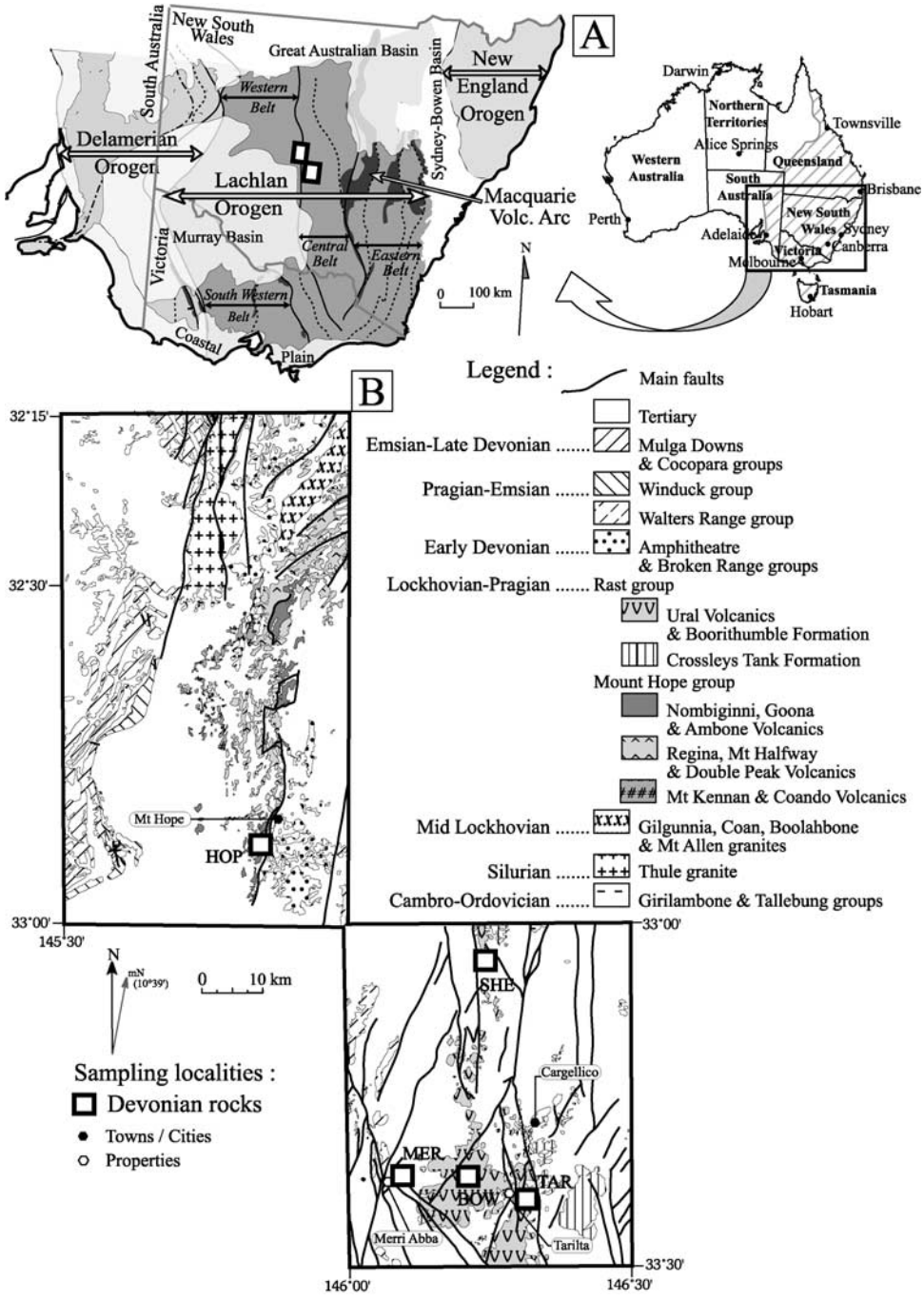


Figure 2. (a) Map of Australia showing the craton (in white) and deformed regions (shaded areas). The studied area is situated in the Tasmanides of New South Wales, southeastern Australia. The Ambone and Ural Volcanics are located just at the boundary between the central and the western subprovinces of the Lachlan Orogen. Redrawn after *Willman et al.* [2002]. (b) Simplified geological map showing the position of the localities targeted; redrawn after *Glen et al.* [1985], *Scheibner* [1985, 1987], *McRae* [1989], and *Colquhoun et al.* [2000].

andesitic and basaltic in small volumes [*Meakin et al.*, 2002]. Radiometric age control is provided by new SHRIMP U/Pb zircon dates, which have been obtained by *Meakin et al.* [2002] for coherent volcanic facies within the Ural Volcanics sequence: 410 ± 8, 414 ± 13, and 413 ± 5 Ma. An older age of 424 ± 6 Ma is interpreted as reflecting inheritance by *Meakin et al.* [2002]. Structurally, the Ural Volcanics show a north to NW trending cleavage, thought to have formed in the late Early to Middle Devonian [*Scheibner and Basden*, 1998], with shallow dips generally to the south or southwest [*Colquhoun et al.*, 2000]. Although the cross section of *Colquhoun et al.* [2000] implies that the Ural Volcanics occupy the core of a regional open syncline, this is not supported by their own outcrop map nor by the

observations of K. Bull, who suggested, on the basis of parallelism of lavas and sills to sedimentary contacts, that no large-scale fold exists (K. Bull, personal communication, 2002). The proposed syncline is based only on the presence of inclined columnar jointing in one outcrop (corresponding to our locality "MER"), but it is well known that columns develop perpendicular to the thermal gradient and not necessarily perpendicular to the paleohorizontal. Albeit the orientation of the columns (whose axes are oriented 304°/56°; number of measurements 51; $\alpha_{95} = 4.79°$), AMS measurements indicate that the Ural Volcanics are flat lying (see section 4) and confirm Kate Bull's detailed observations. Four different localities, labeled "BOW" for Mount Bowen, "TAR" for the Tarilta property, "SHE" for the Shepherds Hill quarry and MER for the Merri Abba Property, have been studied in the Ural Volcanics.

3. Methodology and Sampling

[8] Exposures of unweathered rocks in these regions of low relief are rare. Sample collections have been therefore essentially limited to quarries, which explain the restricted extent of the sampling localities. Samples were drilled using a water-cooled portable drilling machine and orientated using magnetic and sun compass orientations (all directions given herein are corrected for the local declination). Standard samples (2.5 × 2.2 cm) have been cut and measured with 2G cryogenic magnetometer housed in the shielded room of the Ludwig-Maximilians-University of Munich, Germany. Samples have all been thermally demagnetized with a TD-48 furnace and susceptibility routinely measured at every step using a "minikappa" KLF-3 magnetic susceptibility meter inside the shielded room. Anisotropy of magnetic susceptibility (AMS) has been measured with a Kappabridge KLY 2 susceptibility meter (15 positions), and rock magnetism performed with a variable frequency translation balances (VFTB), both operating at the University of Munich.

[9] At locality HOP (latest Pragian Ambone Volcanics near Mount Hope; 145.5473°E/32.8475°S), six sites (30 cores) were collected. In the Ural Volcanics, six sites (29 cores) were obtained in the dacite at Mount Bowen (locality BOW, inheritance age of 424 ± 6 Ma [*Meakin et al.*, 2002]; 146.2155°E/33.3657°S), seven sites (35 cores) in the more rhyolitic rocks of the Shepherds Hill quarry (locality SHE, dated at 410 ± 8 Ma; 146.2469°E/33.0470°S), one site (four strongly altered cores) in a quarry at the Tarilta Property (locality TAR; 146.2432°E/33.4030°S), and six sites (30 cores) in andesitic rocks exposed in a small quarry (~30 meters) at the Merri Abba Property (locality MER, dated at 413 ± 5 Ma; 146.0951°E/33.3649°S).

[10] The quarry at locality TAR contains very altered volcanics, the cause and age of which are unresolved. The results from the AMS and paleomagnetic measurements are unfortunately random and do not provide any reliable information about the alteration event. Data from this locality are therefore not considered further. Concerning the other localities, one specimen from each core has been used for AMS analysis, and sister samples have been subjected to detailed paleomagnetic analysis.

[11] Remagnetizations due to lightning effects can be excluded for all localities, most of which are situated in quarries, as sun compass orientations are in good agreement with those obtained with magnetic compass, and Königsberger ratios [*Königsberger*, 1938] are always low.

[12] However, only locality MER yielded interpretable paleomagnetic results (see section 6, below). For the other localities, paleomagnetic results will be briefly described, but the discussion will mainly bear upon the coherence of the structural control corroborated by AMS measurements.

4. Anisotropy of Magnetic Susceptibility (AMS) and Structural Control

[13] Bedding control is one of the major difficulties in these massive volcanic rocks. Measurements of the AMS indicate the preferred orientation of magnetic minerals, and are used here in two ways. On the one hand, it is one of the best tools to evaluate the degree of internal strain in rocks and is necessary to discard the possibility of deviations of the natural remanent magnetization [e.g., *Tarling and Hrouda*, 1993; *Cogné*, 1987]. On the other hand, the shape and orientation of the ellipsoid of magnetic susceptibility can be regarded as an independent indication to check hypotheses put forward with field observations concerning bedding orientation.

4.1. Ambone Volcanics

[14] For the latest Pragian Ambone Volcanics, at locality HOP, the degree of anisotropy P_J [*Jelinek*, 1981] is weak but well marked ($P_J = 1.05 \pm 0.01$; mean value and standard deviation for the locality), and the shape parameter T [*Jelinek*, 1981; *Hrouda*, 1982] clearly indicates that the ellipsoid of magnetic susceptibility is prolate ($T = -0.44 \pm 0.13$). The directions of the principal axes of magnetic susceptibility of the thirty cores are very well clustered and coincide with the bedding. K_{min} (minimum; shortest axis) and K_{int} (intermediate), and K_{max} (maximum; longest axis) are in horizontal and vertical positions after bedding correction, respectively (Figures 3a and 3b).

4.2. Interpretation

[15] Relationships between bedding and principal axes of magnetic susceptibility mean that the magnetic fabric was acquired prior to folding. However, the position of K_{min} on the horizontal and K_{max} on the vertical is surprising.

[16] The vertical magnetic lineation cannot correspond to the magmatic flow, as these rocks are part of a subhorizontal lava flow (K. Bull, written communication, 2003). The possibility that this vertical elongation results from horizontal contraction associated with the formation of jointing can also be discounted, since magnetite or a magnetite-like mineral, which probably predominantly controlled the susceptibility (see rock magnetism, below), is crystallized at much higher temperature than the temperature at which columnar jointing forms. The vertical elongation must then correspond to tectonic effects. Although horizontal K_{min} usually indicates strong strain, AMS parameters show that this is not the case here as the degree of anisotropy is only about 5%.

[17] The most plausible explanation for the vertical K_{max} is that the ellipsoid of magnetic susceptibility was nearly a sphere, with no original compaction magnetic fabric to overcome. A mild E-W contraction occurred and as these

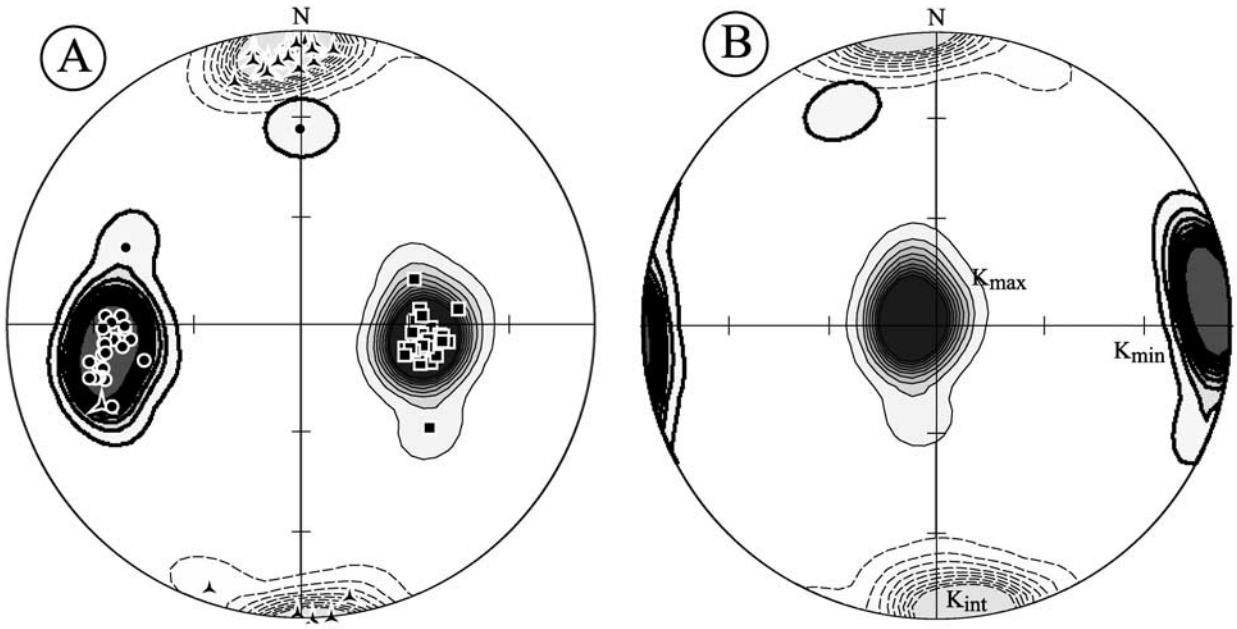


Figure 3. Stereogram (lower hemisphere; equal-area projection) showing the orientation of the principal axes of magnetic susceptibility for the Ambone Volcanics (locality HOP). K_{max} is indicated by squares and thin solid density contours; K_{int} is indicated by triangles and dashed density contours; and K_{min} is indicated by dots and heavy solid density contours. (a) In situ and (b) after bedding correction. For sake of clarity, only density contours are represented after bedding correction. Density contours have been computed using a Gaussian counting model (K = 100).

rocks were close to or at the surface, the material tended to escape with vertical extrusion. This magnetic fabric can be thus the response to a mild E-W contractional event.

[18] Note that AMS measurements were performed again after thermal demagnetization for the localities HOP and MER in order to test the hypothesis that the vertical magnetic lineation resulted from vertical extrusion. Such response could arise from particular magnetic condition where a swap in principal axes of magnetic susceptibility might be observed. Thermal demagnetization, however, does not modify at all the magnetic fabric obtained before heating, and this implies that it results from deformation.

4.3. Ural Volcanics

[19] In the Ural Volcanics at locality BOW, AMS results show that the degree of anisotropy is weak ($P_J = 1.07 \pm 0.06$ for the locality), although the shape parameter T indicates that the ellipsoid of magnetic susceptibility is oblate ($T = -0.26 \pm 0.26$). The magnetic fabric is even closer to isotropic ($P_J = 1.005 \pm 0.002$ and $T = -0.19 \pm 0.33$ for the locality) for the inclined columnar jointing at locality MER. The situation is slightly different for the rhyolitic rocks of locality SHE, where the degree of anisotropy appears more pronounced ($P_J = 1.13 \pm 0.17$ for the locality). The shape of the ellipsoid of magnetic susceptibility remains, however, close to spherical ($T = 0.21 \pm 0.43$), although the mean value point to an oblate shape.

[20] The direction of the poles to the magnetic foliation is approximately horizontal, and approximately vertical for the magnetic lineation in situ, both at localities BOW and MER (Figures 4a and 4b). This is not the case for locality SHE, where the principal axes of magnetic susceptibility are oblique in situ (Figure 4d).

4.4. Interpretation

[21] For these three localities, the magnetic fabric is close to isotropic and internal strain is not strong enough to cause significant deviation of the remanent magnetization. The approximate horizontal and the vertical directions of the principal axes of magnetic susceptibility at localities BOW and MER (Figures 4a and 4b), suggest that no bedding correction is needed as this is consistent with observations of K. Bull (personal communication, 2002). Nevertheless, as in locality HOP, the vertical magnetic lineation is interpreted as due to vertical extrusion, with the horizontal direction reflecting horizontal contraction. The inclined columnar jointing at locality MER led *Colquhoun et al.* [2000] to propose the existence of a regional open syncline for the Ural Volcanics. Rotating the columns to vertical, on the assumption of *Colquhoun et al.* [2000] that they formed perpendicular to bedding, however, rotates the directions of the principal axes of magnetic susceptibility to oblique (Figure 4c) and this is ruled out by the quasi-isotropic nature of the magnetic fabric.

[22] At locality SHE, the situation is different since the directions of the principal axes of magnetic susceptibility, K_{min} and $K_{max,}$ are oblique in situ (Figure 4d). In detail, sites SHE-01, SHE-02, SHE-04, SHE-06, and SHE-07 have well-clustered axes. This clustering means that the common AMS results from sites SHE-01, SHE-02, SHE-04, SHE-06, and SHE-07, postdates formation of the rhyolite dome outlined by magmatic flow foliation observed by K. Bull

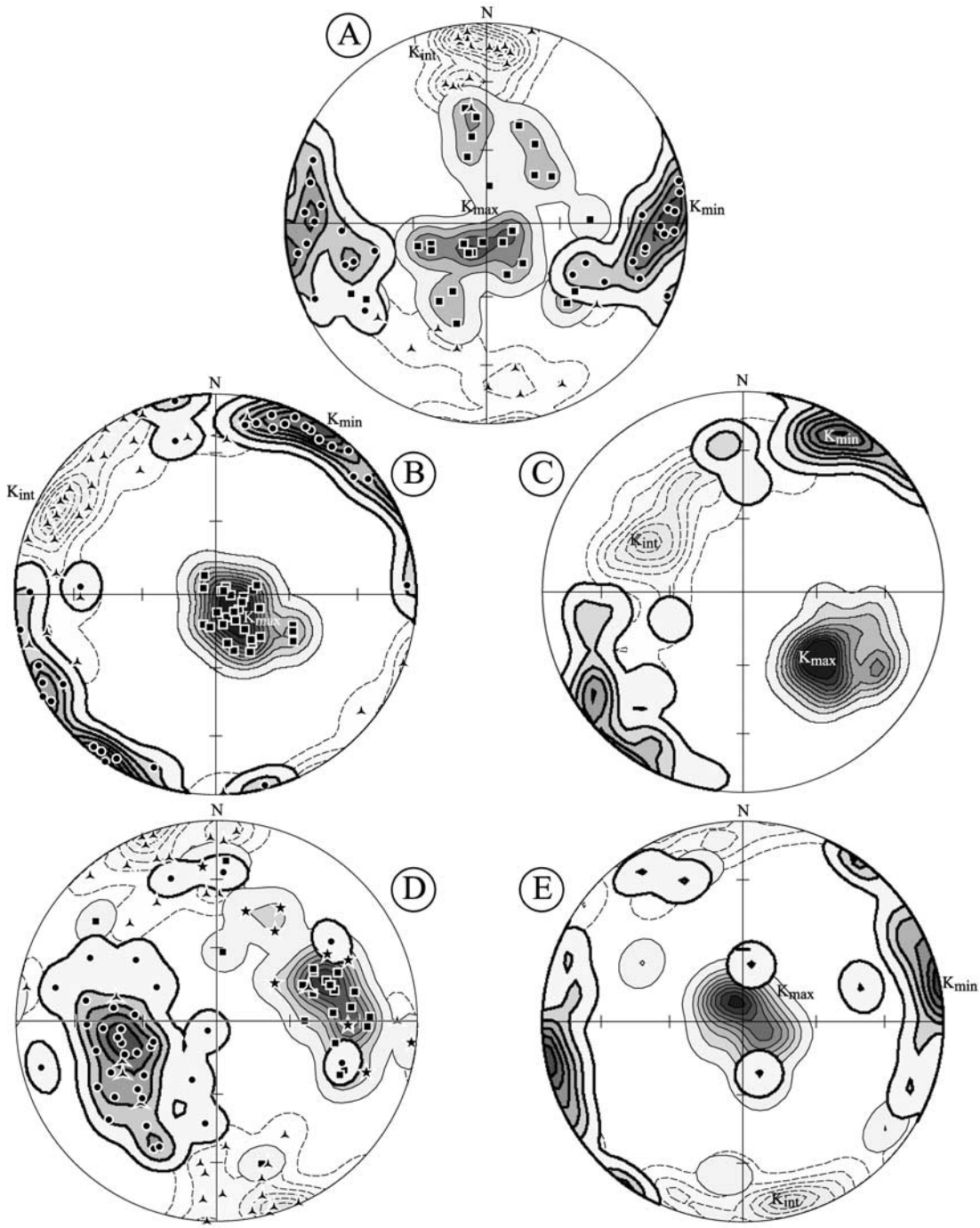


Figure 4. Stereogram (lower hemisphere; equal-area projection) showing the orientation of the principal axes of magnetic susceptibility for the Ural Volcanics. K_{max} is indicated by squares and thin solid density contours (Gaussian model; K = 100); K_{int} is indicated by triangles and dashed density contours; and K_{min} is indicated by dots and heavy solid density contours. (a) For samples from locality BOW represented in situ. (b) For samples from locality MER in situ. (c) After the bedding correction proposed by *Colquhoun et al.* [2000] and consisting in bringing the columnar jointing to the vertical. Note that K_{max} and K_{min} adopt an oblique orientation. (d) For samples from locality SHE in situ. Stars represent K_{max} from sites SHE-03 and SHE-05. They are distributed along a great circle whereas K_{max} from the other sites remain well clustered. (e) Proposed correction for locality SHE consisting in bringing mean K_{max} from all sites but sites SHE-03 and SHE-05 to the vertical and implying local block tilting.

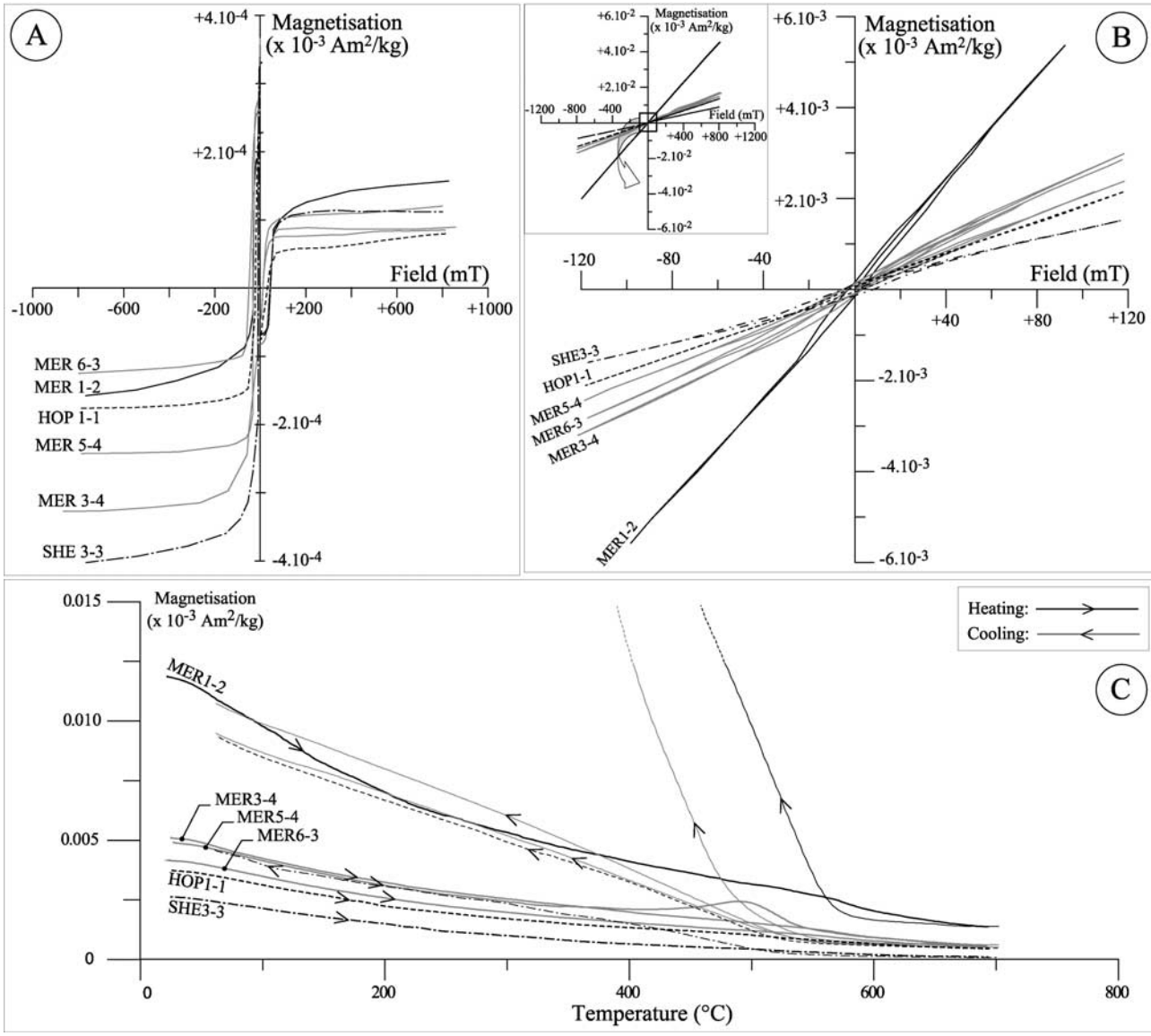


Figure 5. Rock magnetism for specimens from localities MER, HOP, and SHE. (a) Isothemal remanent magnetization (IRM) and backfield curves; (b) hysteresis properties (the inner part of the hysteresis loop (inset) has been enlarged); and (c) thermomagnetic curves (field of 215 mT). Black solid lines indicate specimen from site MER-01 (MER1-2); gray solid lines indicate specimens from other sites at locality MER (MER3-4, MER5-4, MER6-3); black dashed lines indicate specimen from locality HOP (HOP1-1); dotted-dashed lines indicate specimen from locality SHE (SHE3-3).

(personal communication, 2002), and cannot be related to it. In contrast, K_{max} axes from sites SHE-03 and SHE-05 are distributed on a great circle (stars in Figure 4d), probably illustrating nearly equal length between K_{int} and K_{max} axes.

[23] Although no bedding data are available in these massive rocks, the obliquity of magnetic fabric may therefore reflect local block tilting. It is also likely that, as per localities HOP, BOW, and MER, the original shape of the ellipsoid of magnetic susceptibility was close to spherical and the magnetic fabric observed has a tectonic origin.

[24] A tilt correction may consist in bringing the mean K_{max} (calculated using bootstrap analysis [*Tauxe*, 1998], and excluding values from sites SHE-03 and SHE-05) to the vertical (Figure 4e), resulting then in an E-W directed K_{min}. The obtained magnetic fabric becomes therefore consistent with those observed in localities HOP, BOW, and MER, and is in agreement with the paleomagnetic results obtained (see section 6).

5. Rock Magnetic Analysis

[25] Rock magnetic analyses show a notable degree of homogeneity in magnetic mineralogy in the Ambone and Ural volcanic rocks (Figure 5). The main magnetic carrier appears to be low titanium titanomagnetite with Curie temperature ranging in general between 550° and 580°C. The hysteresis parameters witness a strong contribution of paramagnetic material (Figure 5b). The formation of new magnetite, visible on the thermomagnetic curve while cooling, can probably be explained by alteration of paramag-

netic minerals such as phyllosilicates. For some sites, in particular the first site of locality MER (MER-01), which is the closest to the surface of the quarry, thermomagnetic curves show a steeper slope up to 150°–200°C, and the isothermal remanent magnetization and back field curves are not fully saturated, indicating the presence of some oxidized minerals. Moreover, in those cases, the reduced hysteresis loop has a slight wasp-waisted shape, probably related with the coexistence of low- and high-coercivity minerals. Oxidation is interpreted as a result of the strong weathering occurring in Australia (see also section 6).

[26] Results from the Ambone Volcanics point to large grain sizes as they plot close to the pseudosingle to multidomain boundary on the Day plot (H_{CR}/H_C = 4.29 and $M_{RS}/M_S = 7.6 \times 10^{-2}$ [*Day et al.*, 1977]). The presence of relatively large grains of magnetite or titanomagnetite was however expected from such rhyodacite porphyries. In the Ural Volcanics, and in particular at locality MER, these ratios may have lower values (H_{CR}/H_C = 3.39 and $M_{RS}/M_S = 18.06 \times 10^{-2}$) and point more to the pseudosingle domain range. It must be kept in mind, however, that these values are just indication as they reflect the total rock, i.e., probably a mixture of several grain sizes and compositions.

6. Paleomagnetic Results

6.1. Ambone Volcanics

[27] After removal of a viscous or drilling-induced magnetization up to 80°C, two components can be recognized with difficulty. A noisy component C_{PDF} is demagnetized between 80° and ~285°C, and a component C_{HOP} seems to be present up to 575°C, but the magnetic signal is unstable and a viscous magnetization appears from 475°C with usually a strong increase in susceptibility (Figures 6a to 6c). It is not determined yet what carries C_{PDF}, but low titanium titanomagnetite is the main magnetic carrier of C_{HOP}. The orientation of components C_{PDF} corresponds to the present-day field with a direction of magnetization in situ D006°/I-54° (α_{95} = 11.0°/κ = 20.4; κ ratio = 0.9). Orientations of C_{HOP}, however, are too scattered and cannot be considered further. The noisy signal probably results from unstable magnetization associated with the relatively large grain size and weathering effects.

6.2. Ural Volcanics

6.2.1. Localities BOW and SHE

[28] Similarly, a noisy component C_{PDF} is identified between 80° and 185°C in these two localities, corresponding to the present-day field (Figure 6j). In some specimens at locality SHE, a component C_{SHE} can be observed but is lost very rapidly (around 300°C) because the magnetic signal becomes unstable (Figures 6d–6f). It must be noticed, however, that the susceptibility remains stable while demagnetizing, suggesting no laboratory-induced mineralogical changes.

[29] The overall mean direction obtained from single specimen components C_{SHE} from three sites (often with higher unblocking temperatures than in other sites: ~450°C) is D274°/I-47° (N = 12; α_{95} = 8.5°/κ = 27.2) in situ (Figure 6j). Two polarities are present but the reversal test is intermediate and consequently not significant. This direction does not resemble any directions expected for Australia or Gondwana. When corrected for the proposed tilt determined by AMS analysis, which consists of bringing K_{max} to the vertical (Figure 6k), this direction becomes D007°/I-75° (α_{95} = 8.5°/κ = 27.2) and is similar to the midtemperature components determined at Merri Abba (locality MER; see below). Such a direction can correspond to a Carboniferous age of magnetization and it would imply that the local tectonic tilt is younger in age. However, this direction could be only identified in a very limited number of samples at this locality (n = 12; Figures 6g to 6i), and this result must be confirmed before drawing definitive conclusions.

6.2.2. Locality MER

[30] During thermal demagnetization experiments, two paleomagnetic directions can be identified in all thirty samples collected from locality MER (Figure 7). The first, called C_{MT}, is a noisy component demagnetized from 80° to 450°C, i.e., after removal of the viscous or drilling-induced magnetization. The second component, C_{HT}, accounts for approximately 75% of the total remanence and shows a sharp decay in intensity between 525° and 550°C and a corresponding strong increase in susceptibility.

[31] The overall mean direction in situ for C_{MT} (Figures 8a and 8b and Table 1) is D026°/I-60° (α_{95} = 15.2°/κ = 26.3). It has been argued above that no bedding correction is needed, and the confidence cone associated with this direction over-

Figure 6. Examples of paleomagnetic results from rejected sites (localities HOP, BOW, SHE). (a, b, and c) Specimen HOP1-3B; (d, e, and f) specimen SHE3-2B (having an unstable signal above ~200–300°C); and (g, h, and i) specimen SHE3-2B (showing a more coherent signal up to ~550–575°C). Figures 6a, 6d, and 6g show thermal demagnetization curve normalized intensity (J/J_{max}; solid circles) and susceptibility (k/k_{max}; gray circles) versus temperature (in degree Celsius). Boxes correspond to the first derivative (J/T; slope) of the intensity curve. Figures 6b, 6e, and 6h show orthogonal projection [*Zijderveld*, 1967] in geographic coordinates (in situ), where solid circles are projection on the horizontal plan (H) and open circles are projection on the vertical plan (V). Figures 6c, 6f and 6i are stereogram plots of the NRM orientation with thermal demagnetization; solid circles point to the lower hemisphere (positive inclination values), and open circles point to the upper hemisphere (negative inclination values). The star represents the NRM at room temperature (i.e., HOP1-3B: NRM = 3.54×10^{-1} mA/m; SHE3-2B: NRM = 6.526 mA/m; SHE1-2A: NRM = 1.17×10^{-1} mA/m); directions are shown in gray when the signal becomes unstable, Schmidt polar equal-area projection. (j and k) Stereogram for directions of magnetization identified at locality SHE (in situ (Figure 6j) and after the bedding correction proposed after AMS analysis (Figure 6k)). Squares (open, normal; solid, reversed) and gray great circles correspond to single-specimen component C_{PDF}. The overall mean direction (diamond and corresponding α_{95}) is indistinguishable from the present-day dipole direction at the locality (crosses). Open and solid (normal and reversed) circles and solid great circles are single-specimen directions for component C_{SHE}. The overall mean direction (triangle and shaded α_{95}) is consistent with a Carboniferous age of magnetization after the bedding correction inferred from AMS analysis.

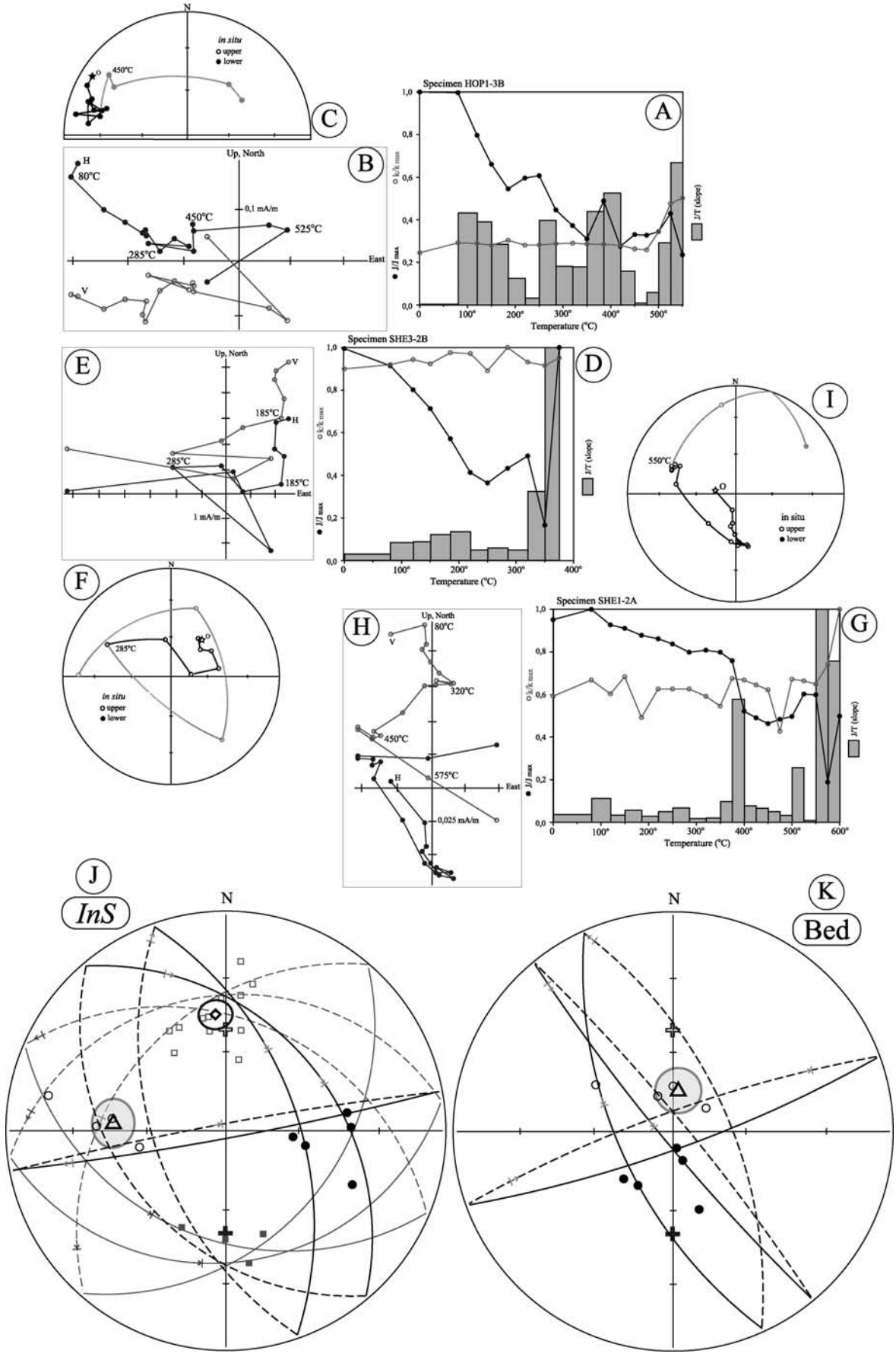

N
in situ
upper
lower
450°C
C
Up, North
H
80°C
450°C
0,1 mA/m
525°C
285°C
East
V
B
Specimen HOP1-3B
A
Temperature (°C)
J/T (slope)
E
185°C
185°C
1 mA/m
Specimen SHE3-2B
D
I
550°C
F
285°C
H
320°C
575°C
0,025 mA/m
Specimen SHE1-2A
G
J
InS
K
Bed

Figure 6

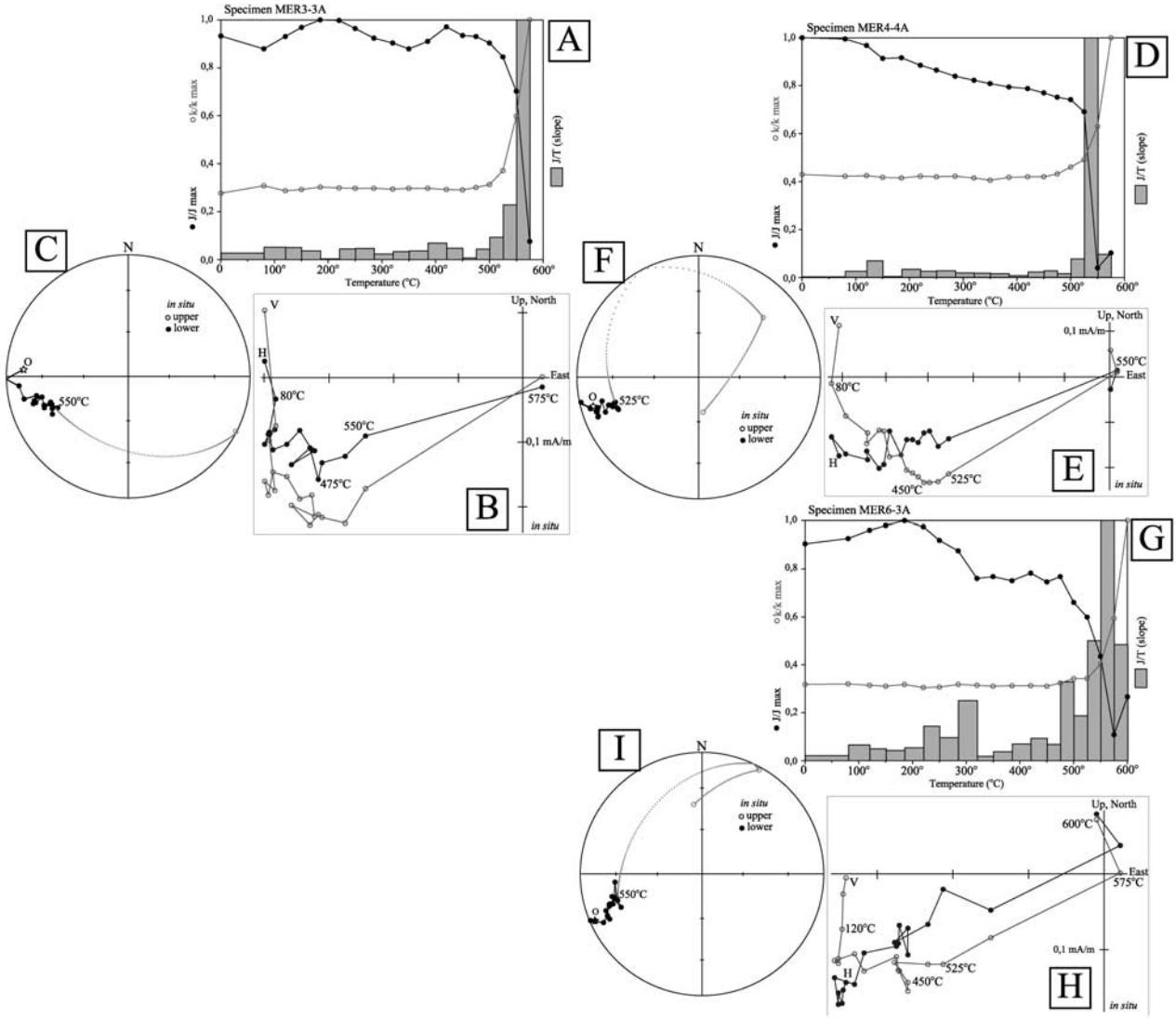


Figure 7. Examples of demagnetization of the Ural Volcanics from the Merri Abba property (locality MER): specimens MER3-3A (Figures 7a, 7b, and 7c), MER4-4A (Figures 7d, 7e, and 7f) and MER6-3A (Figures 7g, 7h, and 7i). Same legend as in Figure 6. Figures 7a, 7d, and 7g show thermal demagnetization curve (J/J_{max}) and susceptibility (k/k_{max}). In Figures 7b, 7e, and 7h orthogonal projection [*Zijderveld*, 1967] is in situ. Figures 7c, 7f, and 7i are stereograms; the star represents the NRM at room temperature; MER3-3A: NRM = 4.17×10^{-1} mA/m; MER4-4A: NRM = 6.46×10^{-1} mA/m; MER6-3A: NRM = 3.70×10^{-1} mA/m; direction are shown in gray when the signal becomes unstable; Schmidt polar equal-area projection.

laps those of the Carboniferous and Early Permian orientations and Late Cretaceous to Cainozoic orientations expected for Australia-Gondwana. However, a Carboniferous age of remagnetization can be favored, because it can be related to the Kanimblan orogenic event that has been best recorded in this region by deformation of late Early–Late Devonian fluviatile sequences [*Glen et al.*, 1996; *Scheibner and Basden*, 1996, 1998; K. Bull, personal communication, 2002].

[32] The overall mean direction for C_{HT} is D252°/I+34° (α_{95} = 7.1°/κ = 88.9) in situ, and again, does not need any bedding correction. One site, MER-01, has a larger confidence cone (shown with dashed line in Figure 8). This site is the closest to the surface in the quarry, and this deviation is likely to be due to weathering effects, as some oxidized minerals are present there (see rock magnetism, above). This means that weathering may have an important influence as deep as 1.5 meters in very coherent rocks, and may be then (partly) responsible for results obtained at Mount Bowen (locality BOW) or Mount Hope (locality HOP) for instance. When site MER-01 is excluded, the site mean directions from the five other sites are very well clustered with a high κ parameter value. The overall mean direction remains however similar to the previous direction based on six sites, since it is D251°/I+31° (α_{95} = 2.2°/κ = 1172.7), in situ. It can be mentioned for information that untilting the columnar jointings (see sections 2 and 4), which axes are oriented 304°/56° (n = 51; α_{95} = 4.79°), would yield a

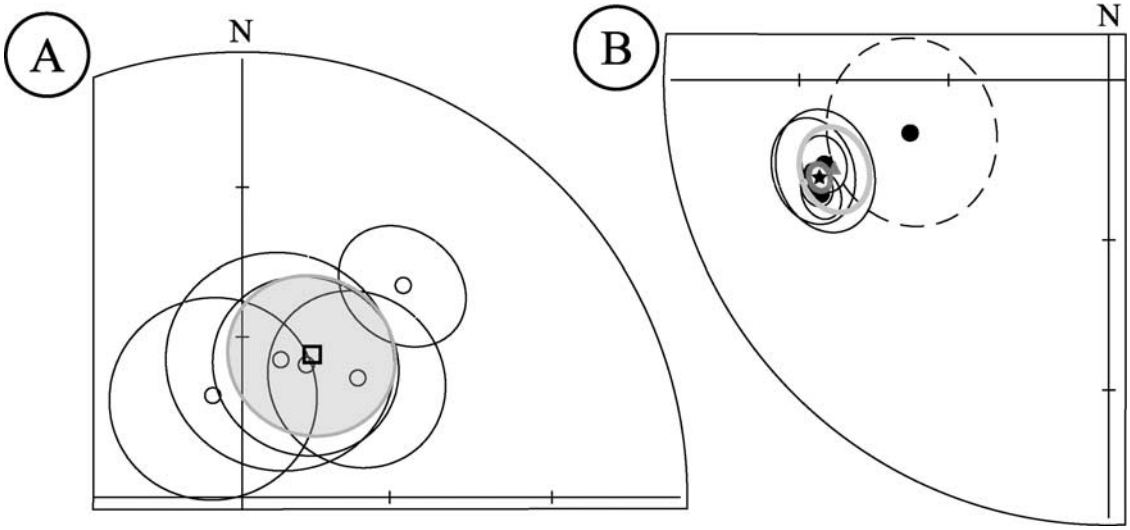


Figure 8. Site mean and overall mean directions of magnetization and associated cones obtained at the 95% confidence level calculated from the 413 ± 5 Ma Ural Volcanics from the Merri Abba property (locality MER). (a) For the midtemperature component in situ, site mean directions, dots (with their corresponding α_{95}) and the overall mean direction, mean mT, square with its shaded α_{95}. (b) High-temperature component in situ, mean HT, shown with a star when site MER-01 is excluded. The direction is, however, very close to the direction obtained with the six sites (gray triangle and α_{95}). Site MER-01 is represented with a dashed line for the confidence cone. Open symbols are negative values (upper hemisphere); solid symbols are positive values (lower hemisphere); Schmidt ''equal''-area projection.

direction D224°/I+31° (α_{95} = 2.2°/κ = 1172.7), and the corresponding pole would not resemble any younger segments of the APW path for Gondwana.

[33] Sampling has been carried out in one lava flow only, where the magnetic mineralogy is rather homogeneous (see section 5). It can be therefore inferred from the high value of the κ parameter (κ = 1172.7) that secular variation has not been averaged out and this direction of magnetization reflects a virtual geomagnetic pole (VGP). It implicitly means also that the magnetization is primary, although no paleomagnetic field test is available. The likelihood that this VGP was recorded during an Earth's field reversal is not null but very small. Hence, if a 30° cone corresponding to the largest plausible deviation of a VGP (see, e.g., *Soffel* [1991] or *Collinson* [1983]) to a specific pole position is drawn (dashed line in Figure 9b), the position corresponding to this direction (Table 2) is compatible with an earliest Devonian position suggested by the X type of the APW path for Gondwana but not the Y-type path (Figure 9b). Moreover, an overprint should be consistent with any younger segments of the APW path for Gondwana-Australia (disregarding the 30° cone, since the pole should not be considered as a VGP anymore), which is not the case.

7. Discussion

[34] Owing to the paucity of Early Devonian data in the paleomagnetic database (IAGA paleomagnetic database [*McElhinny and Lock*, 1996]; online version 4.5, 2003), the VGP is compared to the 31 paleomagnetic data (Table 3) ranging in age from Late Silurian to Middle Devonian [425–370 Ma] that exist for Gondwana. These poles are very scattered, and therefore include erroneous data that must be filtered. *Van der Voo* [1988, 1993, p. 67] proposed seven criteria to assess the quality of paleomagnetic data. The first three criteria (1) ''A well-determined rock age and a presumption that magnetization is the same age''; (2) ''Sufficient number of samples'' such as $N \geq 24$, $\kappa \geq 10$, and $\alpha_{95} \leq 16°$; and (3) ''Adequate demagnetization that demonstrably includes vectors subtraction'' (i.e., ''Demag-Code'' ≥ 3 in the IAGA paleomagnetic database) are regarded as fundamental, because data failing can be considered as statistically unreliable. Since they are objective, any doubt on these criteria renders a result questionable. The four additional criteria are very important in assessing data quality, but they are more subjective and their absence does not necessarily imply that results are unreliable. The first three criteria have thus been used to determine statistically reliable data, and only nine results satisfy them (indicated in Table 3; Figure 9a). Among them however, three data still appear erroneous. The Hazzel Matti Formation pole from Algeria (pole 3 in Table 3) and the Gneiguira Supergroup pole from Mauritania (pole 4) closely resemble the Carboniferous–Permian segments of the APW paths for Gondwana, and are likely to be overprinted (Figure 9a). The pole obtained from the Ravenswood Batholith and its aureole in far north Queensland (pole 24) also lies far from any expected position for Gondwana (Figure 9a), and probably reflects block rotation and/or tilting associated with Late Carboniferous–?Triassic deformation. Two groups of poles remains: the first supporting a direct (X) path and the second supporting a Silurian–Devonian loop (Y path). All poles supporting the Y type, with the exception of one, are obtained from rocks situated

Table 1. Site Mean and Overall Mean Directions Obtained From the Ural Volcanics at the Merri Abba Property[a]

Sites	n/N	D	I	**R**	α_{95}	κ
Midtemperature Components (B = 5)						
MER-01	4/5	037.2	−39.1	4.0	11.4	66.2
MER-02	5/5	044.1	−58.9	4.8	16.5	22.6
MER-03	5/5	025.8	−62.5	4.8	17.0	21.3
MER-05	3/5	015.6	−63.3	2.9	20.8	36.3
MER-06	4/5	343.8	−70.3	3.9	19.1	24.2
Mean mT	21/30	026.0	−60.2	4.8	15.2	26.3
High-Temperature Components (B = 6)						
MER-01	5/5	255.0	+51.2	4.8	16.6	22.1
MER-02	5/5	252.2	+32.2	4.9	10.2	60.7
MER-03	5/5	253.5	+32.7	5.0	4.7	267.4
MER-04	5/5	250.0	+29.6	5.0	3.3	529.7
MER-05	5/5	248.3	+30.0	5.0	4.1	350.0
MER-06	5/5	252.8	+29.9	4.9	8.7	78.2
Mean HT	30/30	251.8	+34.3	5.9	7.1	88.9
High-Temperature Components (B = 5; When Site MER-01 Is Excluded)						
Mean HT	25/30	251.3	+30.9	5.0	2.2	1172.7

[a]Locality "MER" 413 ± 8 Ma; n, number of samples used out of N, total number of samples collected; B, number of sites; α_{95}, cone calculated at the 95% confidence level; κ, precision parameter [*Fisher*, 1953]; **R**, resultant vector; *D*/*I*, declination/inclination in situ (geographic coordinates, since no bedding correction is needed).

east of the Macquarie Volcanic Arc. The Air Intrusive pole (pole 6 in Table 3) is the only result from Africa in accordance with the loop. It has been argued, nevertheless, that this pole obtained in Niger is affected by a Carboniferous overprint, related to the formation of Pangea, much like the Hazzel Matti and the Gneiguira poles [e.g., *Bachtadse and Briden*, 1991].

[35] The pole obtained at locality MER points off the paths for Gondwana, but given that the result presented here is believed to represent a VGP, it is likely that the corresponding paleopole position lies within a 30° cone around the VGP. Consequently, this result supports the X type of the APW path for Gondwana, and suggests that the proposed Siluro-Devonian loop passing by Chile for the APW path for Gondwana (Y path in Figures 1 and 7) may be an artefact due to block rotations in the eastern Lachlan Orogen.

[36] An alternative explanation to the discrepancy between the VGP position and the Siluro-Devonian segments of the paths, however, could be block rotation of the study area. A clockwise block rotation of approximately 20 degrees can bring the pole to the X-type APW path, whereas a ∼105° is necessary to match the Y-type APW path (Figure 9b). Nevertheless, there is no field evidence to support the hypothesis of block rotation in the area of study. The hypothesis of deviation of the VGP relative to the paleopole is

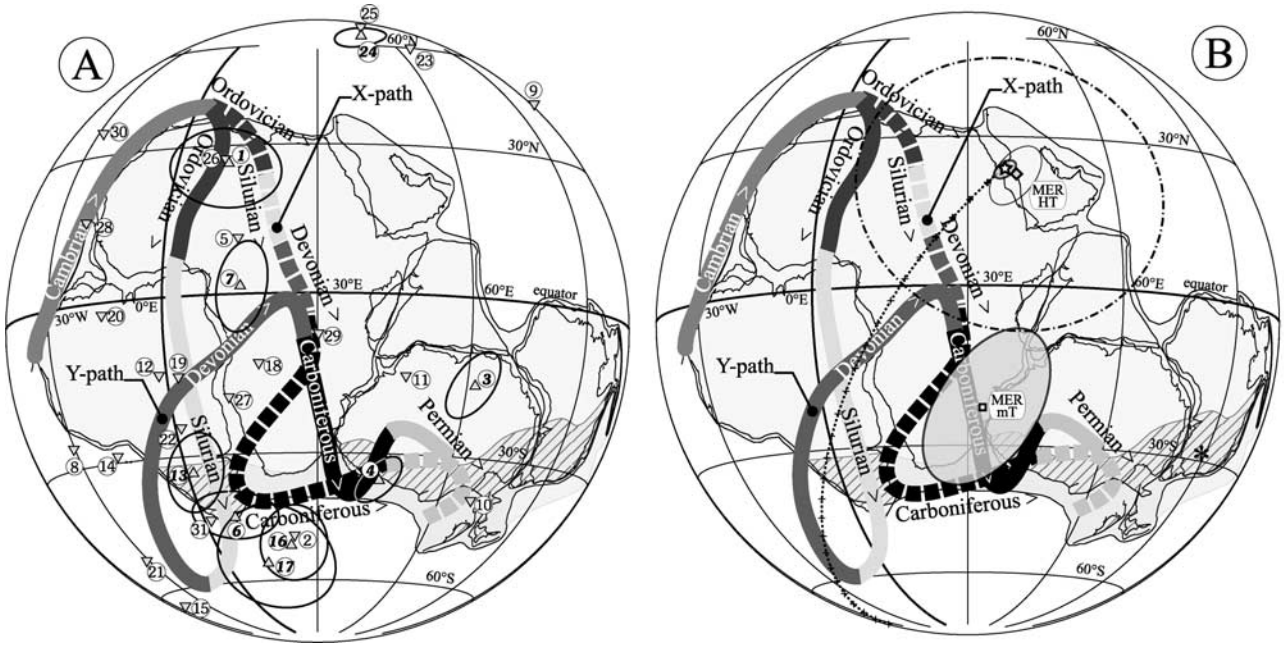


Figure 9. (a) Silurian to Middle Devonian poles from the IAGA paleomagnetic database (version 4.5) listed in Table 3 (same numbers). Triangles with their associated solid confidence cones (numbers in bold) refers to poles satisfying the first Van der Voo's criteria (see text), whereas the other (inverted triangles without confidence cones) do not. (b) Corresponding paleopole for the midtemperature component (MER mT) is shown with a square and shaded confidence ellipse. The high-temperature component (MER HT) is believed to represent a VGP and is shown with a star and shaded confidence ellipse when based on five sites (MER-01 excluded). Note that the corresponding Early Devonian (413 ± 5 Ma) paleopole is likely to lie in a 30° cone (dashed circle) drawn around the VGP. For information, the pole based on the six sites is also shown: diamond and thin confidence ellipse. The consequence of block rotation is represented by the dashed small circle with crosses every 5°, centered on a vertical axis positioned at locality MER (asterisk). It is plotted on a reconstruction of Gondwana in African coordinates as presented in Figure 1.

Table 2. Paleopole and VGP Obtained From the Ural Volcanics at the Merri Abba Property[a]

		Australian Coordinates			African Coordinates						
B	n/N	Plat	Plong	Paleo-colatitude	Plat	PLong	α_{95}	**R**	κ	dp	dm
				Mean Values for the Midtemperature Components, MER-mT							
5	21/30	−67.9	084.4	−48.8	−22.6	032.6	15.20	4.8	26.3	17.5	23.1
				Mean Values for the High-Temperature Components, MER-HT							
6	30/30	−25.1	062.9	+71.2	+21.7	039.6	7.14	5.9	88.9	4.7	8.2
5	25/30	−24.4	060.6	+73.3	+23.0	037.7	2.24	5.0	1172.7	1.4	2.5

[a]B, number of sites; n, number of samples used out of N, total number of samples collected; PLong/Plat, paleopole longitude/latitude in Australian and in African coordinates, using the rotation parameters of *Lawver and Scotese* [1987]; α_{95}, cone calculated at the 95% confidence level; κ, precision parameter [*Fisher*, 1953]; **R**, resultant vector; dp/dm, semiaxis of the ellipse of confidence.

Table 3. Poles [425–370 Ma] for Gondwana From the IAGA Paleomagnetic Database[a]

Pole	Ref	Name, Country	Demag Code	N	κ	α_{95}	PLong	Plat	Slong	Slat	Tests	Age	Authors
						Africa							
1[b]	2189	Gilif Hills Volc-1, Sudan	4	39	18.7	10.8	191.6	−25.9	032,7	+17.8	R−	377[R]	*Bachtadse and Briden* [1991]
2	2189	Gilif Hills Volc-2, Sudan	4	22	48.1	7.0	203.5	+48.8	032,7	+17.8	Rc	377[R]	*Bachtadse and Briden* [1991]
3[b]	2884	Hazzel Matti Fm, Algeria	4	53	378.0	6.3	241.7	+16.2	002,1	+24.9	F+	379[F]	*Smith et al.* [1994]
4[b]	1761	Gneiguira SGp, Mauritania	3	44	89.0	5.1	224.0	+35.5	347,7	+18.1		397[S]	*Kent et al.* [1984]
5	1416	Bokkeveld Gp, South Africa	4	13	33.0	7.0	015.0	+10.0	019,0	−33.5		399[S]	*Bachtadse et al.* [1987]
6[b]	1364	Air Intrusive, Niger	4	53	44.7	6.6	188.6	+43.4	008,8	+19.3		407[R]	*Hargraves et al.* [1987]
						Australia							
7[b]	1345	Canning Basin, WA	4	89	61.6	7.8	218.0	+49.1	125,6	−18.3		370[F]	*Hurley and Van der Voo* [1987]
8	0805	Catombal Gp, NSW[c]	2	66	33.0	10.0	132.0	+49.0	148,6	−32.3	F+	370[S]	*Williamson and Robertson* [1976]
9	1045	Lochiel Fm, NSW[c]	3	25	18.1	18.5	092.0	+28.0	149,8	−36.9		370[S]	*Embleton and Shepherd* [1977]
10	2911	Ellenborough Basalt, NSW[c]	4	11	–	11.4	326.2	+49.4	152,5	−31.5		370[S]	*Schmidt et al.* [1994]
11	1046	Mulga Downs Gp, NSW[c]	2	12	17.7	8.0	276.0	+54.0	143,5	−32.8		374[F]	*Embleton* [1977]
12	2574	Hermannsburg Ss, NT	4	24	35.6	21.0	180.9	+61.0	133,0	−24.0		374[F]	*Li et al.* [1991]
13[b]	1565	Comerong Volc, NSW[c]	4	52	38.8	7.4	150.7	+76.9	150,0	−35.5		375[F,R]	*Schmidt et al.* [1986]
14	2574	Parke Siltstone, NT	4	9	12.9	14.9	138.1	+60.9	132,1	−23.6		384[F]	*Li et al.* [1991]
15	1612	Wellington-Cowra R., NSW[c]*	4	42	10.0	30.6	071.8	+42.2	149,0	−33.6		397[S]	*Goleby* [1980]
16[b]	3262	Mt Leyshon dykes, NSW[c]	4	25	46.1	7.6	018.8	+78.0	146,5	−20.0	C+	399[R]	*Clark* [1996]
17[b]	1365	Snowy River Volc, Va[c]	4	47	25.6	9.7	042.7	+74.3	148,3	−37.4	F+	404[F]	*Schmidt et al.* [1987]
18	0182	Bowning Gp Volc, NSW[c]	2	25	36.8	10.0	225.0	+64.0	148,8	−34.5	G+	415[F]	*Luck* [1973]
19	0278	Mugga Mugga R., ACT[c]	2	17	29.0	7.0	187.9	+63.2	149,4	−35.1		422[S]	*Briden* [1966]
20	1612	Cowra-Molong R., NSW[c]	4	154	29.3	17.3	174.0	+44.5	149,0	−33.6	F+	423[S]	*Goleby* [1980]
21	0182	Laidlaw-Douro Volc, NSW[c]	2	35	18.4	10.0	091.0	+54.0	148,9	−34.7	Rc	424[F]	*Luck* [1973]
22	0182	Ainslie Volc, ACT[c]	2	24	33.4	11.0	173.0	+71.0	149,0	−35.0		425[S]	*Luck* [1973]
23	2574	Mereenie Ss, NT	4	51	7.8	25.6	062.7	+15.7	132,0	−24.0		425[S]	*Li et al.* [1991]
24[b]	3262	Ravenswood R., Qld[c]	4	51	20.0	7.1	232.7	−17.5	146,4	−20.1		425[R]	*Clark* [1996]
25	3262	Mt Leyshon dykes, Qld[c]	4	14	39.1	8.3	231.9	−21.7	146,4	−20.1		425[R]	*Clark* [1996]
						South America							
26	0283	Salta-Jujuy red beds, Argentina 2	10	–	8.0	338.0	+09.0	294,0	−23.0			390[S]	*Creer* [1965]
27	0613	Picos-Passagem Series, Brazil	2	12	–	32.0	133.0	+30.0	317,0	−06.0	R−	397[F]	*Creer* [1970]
28	0613	Sediments, Bolivia	2	26	–	22.0	307.0	+17.0	293,0	−17.0		404[S]	*Creer* [1970]
29	1159	Nunorco Fm Gp A, Argentina	3	12	85.7	10.0	157.0	+30.0	291,9	−27.8		407[S,R]	*Valencio et al.* [1980]
30	1159	Nunorco Fm Gp B, Argentina	3	5	308.2	14.3	316.0	+36.0	291,9	−27.8		407[S,R]	*Valencio et al.* [1980]
31	2639	Herrada Member, Argentina	3	10	23.0	10.3	103.5	+42.0	294,6	−41.6	F+	407[F]	*Rapalini and Vilas* [1991]

[a]Ref, reference number given in the IAGA paleomagnetic database (*McElhinny and Lock* [1996]; on-line version 4.5, 2003; see also detailed references therein); Name, Volc, volcanics; Fm, formation; Gp/SGp, group/supergroup; Ss, sandstone; R, rocks; for Australia, WA, Western Australia; NT, Northern Territories; Qld, Queensland; NSW. New South Wales; ACT, Canberra A.C.T.; Va, Victoria; DemagCode, code used in the database and assessing the quality of the demagnetizing process; N, number of samples; κ, precision parameter [*Fisher*, 1953]; α_{95}, cone calculated at the 95% confidence level; PLong, Plat, paleopole longitude/latitude; SLong, Slat, longitude/latitude of the study area; tests, paleomagnetic tests (R, reversal test; F, fold test; C, contact test; G, conglomerate test; plus and minus mean positive & negative tests, and R_C corresponds to a positive reversal test of class C); Age, mean age calculated from the age window given in the database (R, radiometric (R-Sr, U-Pb, K-Ar); F, fossils; S, stratigraphic).

[b]Data satisfy the first Van der Voo's criteria (see text).

[c]Results coming from eastern Australia, in particular from the Lachlan Orogen.

therefore preferred, and the area of study in the central subprovince of the Lachlan Orogen can be regarded as fixed relative to the craton of Gondwana in the Early Devonian.

[37] The closest equivalent in age to the VGP from locality MER and the most reliable paleomagnetic data from the Lachlan Orogen with its convincing positive fold test is the result from the Siluro-Devonian Snowy River Volcanics (SRV pole [*Schmidt et al.*, 1987]; pole 17 in Table 3). Assuming that the X APW path of *Bachtadse and Briden* [1991] is the reference path, the SRV pole requires both terrane displacement ''northward'' (looking at present-day configuration) and counterclockwise rotation (~60°) around a vertical axis at the locality of study to match the Late Silurian–Early Devonian segment of the X path. Another good paleomagnetic result from the eastern Lachlan Orogen is the pole obtained from the Middle Devonian Comerong Volcanics [*Schmidt et al.*, 1986] (pole 13 in Table 3). This pole can match the X APW path of *Bachtadse and Briden* [1991] simply by rotation (~45/50°). Such a rotation could have occurred in the Early Carboniferous Kanimblan orogenic event, characterized by contractional and strike-slip faulting [*Glen*, 2005]. While an X type of APW path is favored, it is still too ill defined at present to actually assess the amount of displacement and rotation. Any palinspastic reconstruction of eastern Australia based on these data would be thus over interpreted.

[38] An X type of APW path is in addition in better agreement with lithological and paleoclimatic indicators as shown for instance by *Scotese et al.* [1999]. The Early Devonian pole position deduced from their calculation lies more to the southwest than the one suggested by the X APW path of *Bachtadse and Briden* [1991] and by locality MER, but paleoclimatic indicators can be viewed as ''strong filter'' of the paleolatitudes (a warm zone is broad and easily encompass 30° in latitude). For comparison, *Scotese et al.* [1999] indicate the presence of evaporite and bauxite very close to the South Pole in the Cretaceous. Nevertheless, they also favor an X-type APW path for Gondwana.

8. Conclusions

[39] Results from AMS measurements in the Ambone Volcanics indicate a good correspondence between the magnetic fabric and the bedding orientation. Results from the Ural Volcanics confirm that these rocks are flat lying as observed by K. Bull (personal communication, 2002), although it appears that local block tilting occurred at locality SHE, possibly in the Early Carboniferous. The regional magnetic fabric is unusual with a magnetic lineation (K_{max}) in vertical position. It is interpreted as a response of minor vertical elongation during E-W shortening.

[40] Reliable paleomagnetic results could be obtained only from the andesitic polygonal columns of the Ural Volcanics exposed in the small quarry on the Merri Abba Property (146.0951°E/33.3649°S). These rocks are well dated at 413 ± 5 Ma by the U/Pb zircon SHRIMP technique [*Meakin et al.*, 2002]. The high-temperature component is clearly determined at this locality and corresponds to a pole position of $\lambda = 23.0°N/\phi = 037.7°E$ (dp = 1.4°/dm = 2.5°; in African coordinates) when site MER-01 is excluded (Figure 9b). This does not correspond to any expected paleopole of younger ages.

[41] Whether the area of study is fixed or not, there is now increasing evidence that part of the Central and/or the eastern subprovinces of the Lachlan Orogen have undergone rotation, and the two solutions lead to the same conclusion: the Lachlan Orogen as a whole was not stable until at least the mid-Paleozoic. The use of paleomagnetic data from southeastern Australia should be therefore avoided in construction of an APW path for Gondwana. Rejecting these suspect data leads to a rather direct APW path through Africa.

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